Example 5.7

Calculate the geometric mean of the annual percentage growth rate of profits in business corporate from the year 2000 to 2005 is given below

50, 72, 54, 82, 93

Solution:

x_i	50	72	54	82	93	**Total**
log x_i	1.6990	1.8573	1.7324	1.9138	1.9685	**9.1710**

$$\text{G.M.} = \text{Antilog}\ \frac{\sum_{i=l}^{n} \log x_i}{n}$$

$$= \text{Antilog}\ \frac{9.1710}{5}$$

$$= \text{Antilog}\ 1.8342$$

$$\text{G. M.} = 68.26$$

Geometrical mean of annual percentage growth rate of profits is 68.26

Example 5.8

The population in a city increased at the rate of 15% and 25% for two successive years. In the next year it decreased at the rate of 5%. Find the average rate of growt

Solution:

Let us assume that the population is 100

Percentage rise in population	**Population at the end of year** x_i	**log** x_i
15	115	2.0607
25	125	2.0969
5	95	1.9777
		6.1353

$$\text{G.M} = \text{Antilog}\ \frac{\sum_{n}^{n} \log x_i}{n}$$

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